

4.8.3 Identification of ions by chemical and spectroscopic means (chemistry only)

4.8.3.1 Flame tests

Content	Key opportunities for skills development
Flame tests can be used to identify some metal ions (cations). Lithium, sodium, potassium, calcium and copper compounds produce distinctive colours in flame tests: • lithium compounds result in a crimson flame • sodium compounds result in a yellow flame • potassium compounds result in a lilac flame • calcium compounds result in an orange-red flame • copper compounds result in a green flame. If a sample containing a mixture of ions is used some flame colours can be masked.	AT 8 An opportunity to investigate flame colours.
Students should be able to identify species from the results of the tests in 4.8.3.1 to 4.8.3.5. Flame colours of other metal ions are not required knowledge.	WS 2.2

4.8.3.2 Metal hydroxides

Content	Key opportunities for skills development
Sodium hydroxide solution can be used to identify some metal ions (cations). Solutions of aluminium, calcium and magnesium ions form white precipitates when sodium hydroxide solution is added but only the aluminium hydroxide precipitate dissolves in excess sodium hydroxide solution. Solutions of copper(II), iron(II) and iron(III) ions form coloured precipitates when sodium hydroxide solution is added. Copper(II) forms a blue precipitate, iron(II) a green precipitate and iron(III) a brown precipitate.	AT 8 An opportunity to make precipitates of metal hydroxides.
Students should be able to write balanced equations for the reactions to produce the insoluble hydroxides. Students are not expected to write equations for the production of sodium aluminate.	WS 2.2

4.8.3.3 Carbonates

Content	Key opportunities for skills development
Carbonates react with dilute acids to form carbon dioxide gas. Carbon dioxide can be identified with limewater.	

4.8.3.4 Halides

Content	Key opportunities for skills development
Halide ions in solution produce precipitates with silver nitrate solution in the presence of dilute nitric acid. Silver chloride is white, silver bromide is cream and silver iodide is yellow.	

4.8.3.5 Sulfates

Content	Key opportunities for skills development
Sulfate ions in solution produce a white precipitate with barium chloride solution in the presence of dilute hydrochloric acid.	

Required practical 7: use of chemical tests to identify the ions in unknown single ionic compounds covering the ions from sections [Flame tests](#) (page 73) to [Sulfates](#) (page 74).

AT skills covered by this practical activity: 1 and 8.

This practical activity also provides opportunities to develop WS and MS. Details of all skills are given in [Key opportunities for skills development](#) (page 107).

4.8.3.6 Instrumental methods

Content	Key opportunities for skills development
Elements and compounds can be detected and identified using instrumental methods. Instrumental methods are accurate, sensitive and rapid. Students should be able to state advantages of instrumental methods compared with the chemical tests in this specification.	

4.8.3.7 Flame emission spectroscopy

Content	Key opportunities for skills development
Flame emission spectroscopy is an example of an instrumental method used to analyse metal ions in solutions. The sample is put into a flame and the light given out is passed through a spectroscope. The output is a line spectrum that can be analysed to identify the metal ions in the solution and measure their concentrations.	AT 8 An opportunity to observe flame spectra using a hand-held spectroscope.
Students should be able to interpret an instrumental result given appropriate data in chart or tabular form, when accompanied by a reference set in the same form, limited to flame emission spectroscopy.	WS 3.6 MS 4a